

TECHNICAL CORRECTION PACKAGE

Lanmodulin Dy-selectivity workflow

Corrected figures, tables, reviewer-response map, and repository remediation

Manuscript: JMGM-D-26-01758
Scope: Technical correction only; main manuscript intentionally unchanged
Repository: <https://github.com/RahatulAshakin/lanm-dy-selectivity>
Audited base: 549b22918e1edf409e0c75831cb9af7069468275
Prepared: 2026-08-16

Targeted result after correction

The technically supportable result is a model-prioritized Hans pocket candidate for additional quantum and experimental testing. The present repository does not establish thermodynamic binding free energy, Dy selectivity, zero-waste separation, or a validated quantum-state mechanism.

Area	Status	Submission use
Classical data correction	Ready	Corrected masks, funnel, lineage, model settings, OpenMM replica data, and dimensionless model score
Quantum evidence	Not ready	Raw ORCA/CASSCF/CASPT2 files are absent; Figures 6-7 cannot be regenerated defensibly
Manuscript text	Unchanged	Author name, title/abstract claims, citations, and prose edits are recorded but not applied
Repository remediation	Prepared	New technical-correction artifacts, reproducible builder, provenance tables, and corrected README notice

Status is based on repository audit at main commit 549b22918e1edf409e0c75831cb9af7069468275.

1. Scope and correction decision

This package responds to the supplied reviewer comments through technical and reproducibility corrections only. It does not alter the manuscript. The package separates supported corrections from author-dependent work so that unsupported quantum values, mechanisms, or experimental claims are not introduced merely to satisfy a review point.

1.1 What can be corrected now

Correction	Concrete update
Metric terminology	Remove the energy-like transformation and report a dimensionless Composite Quality Index.
Design accounting	Reconcile 70 role memberships to 66 unique positions and separate positions from generated sequences.
Sampling transparency	Report exact MPNN temperatures, seeds, counts, and shortlist units.
Rosetta description	Replace structural-realism wording with one-structure score-only triage and explicit limitations.
OpenMM reporting	Expose all 45 replica/metal/candidate means, parameter values, seeds, duration, metric definition, and force-field caveats.
Structure scope	Distinguish deposited A-D chains from author/PISA A2 assemblies and clarify chain-A versus multichain computational contexts.

Full machine-readable tables are included in the `tables/` directory.

1.2 What cannot be corrected without author data

The repository contains no ORCA inputs or outputs and no files named for CASSCF or CASPT2. Therefore natural occupations, active-space root character, CASPT2 settings, intruder-state diagnostics, cluster sizes, relativistic treatment, and spin-orbit coupling cannot be reconstructed. Existing quantum Figures 6-7 should not be submitted as regenerated figures until the source files listed in Section 6 are deposited.

2. Reviewer-response correction matrix

The matrix below is the shareable point-by-point technical response. 'Blocked' means the correction requires source data that are not in the repository; it does not mean a value should be estimated.

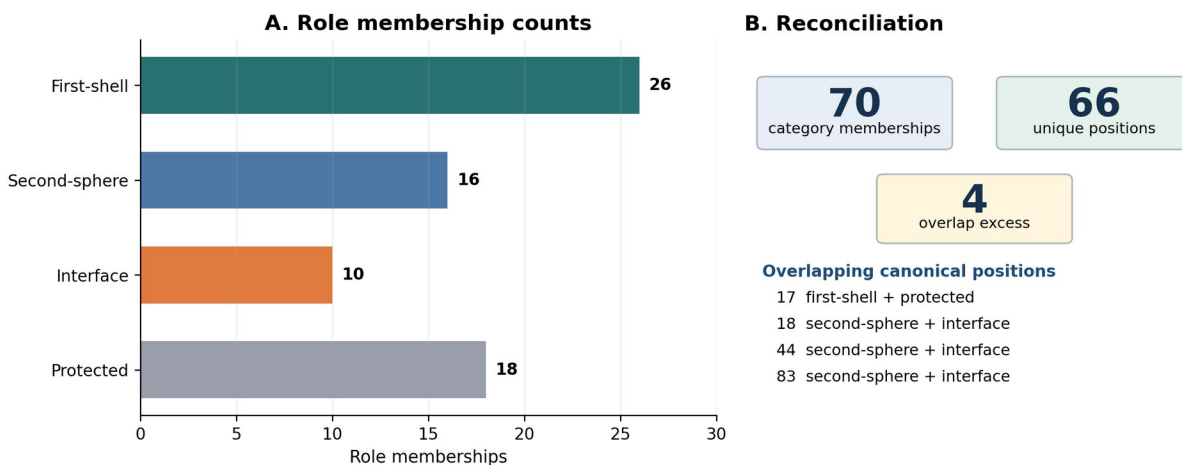
Item	Reviewer issue	Technical correction	Status
R1.1 / R2-B	Undefined and thermodynamically invalid pocket delta-G proxy	Replace with dimensionless Composite Quality Index = 0.7 ligand confidence + 0.3 overall confidence; remove -RT ln transformation. Evidence: Figure TC5; model_quality_index.csv.	Corrected
R1.2	Natural occupations and active-space suitability	Do not regenerate without raw multiroot orbital and occupation data. Require all roots, orbitals, occupations, and active-space justification. Evidence: quantum_evidence_gap_register.csv.	Blocked: source data absent
R1.3	CASPT2 shift, intruder states, number of states	Require program/version, state averaging, imaginary/IPEA shifts, state counts, convergence, and intruder-state diagnostics. Evidence: quantum_evidence_gap_register.csv.	Blocked: source data absent
R1.4	Reduced-cluster size and environmental effect	Require atom/residue inventory, total charge/multiplicity, caps/constraints, coordinates, and sensitivity comparison. Evidence: quantum_evidence_gap_register.csv.	Blocked: source data absent

Item	Reviewer issue	Technical correction	Status
R1.5	Zero-waste claim lacks experimental evidence	Technical package states that no experimental separation or waste-balance evidence is present and narrows the result to a computational candidate-prioritization claim. Evidence: Technical report Sections 1 and 6.	Flagged; manuscript wording outside scope
R1.6	Only Al and Fe competitors	State that Al(III)/Fe(III) were the tested off-target subset. Do not generalize to Ca/Mg/Zn; recommend an expanded panel. Evidence: modeling_specification.csv.	Corrected limitation
R1.7	MPNN parameters and filtering unspecified	Expose exact temperatures, seeds, generated/unique/retained counts, and selection units. Evidence: Figure TC2; sequence_funnel.csv; modeling_specification.csv.	Corrected
R1.8 / R2-S4	Rosetta realism criteria/cutoffs unspecified	Correct description to score-only triage: score_jd2, nstruct=1, fixed seed, no relax, no acceptance cutoff, within-topology ranking only. Evidence: modeling_specification.csv.	Corrected and qualified
R1.9	Hans pocket versus interface unclear	Distinguish chain-A pocket model from chain-A-designed/B-D-fixed crystallographic context; attribute score difference to model confidence, not inferred structural causality. Evidence: Figure TC3; finalist_lineage.csv.	Corrected
R2-A1	Spin-orbit coupling not documented	Require SOC/RASSI-SO/NEVPT2-SO treatment or an explicit scope-limited justification. No claim made from absent data. Evidence: quantum_evidence_gap_register.csv.	Blocked: source data absent
R2-A2	Relativistic Hamiltonian and basis sets absent	Require exact Hamiltonian/ECP, basis sets for all atoms, auxiliary bases, scalar-relativistic settings, and software version. Evidence: quantum_evidence_gap_register.csv.	Blocked: source data absent
R2-A3	Active orbitals/occupations across roots not demonstrated	Require orbital plots and natural occupation numbers for every reported root; verify seven 4f orbitals remain active. Evidence: quantum_evidence_gap_register.csv.	Blocked: source data absent
R2-C1	Metal force-field provenance and Al/Fe artifacts	Report actual generic OPC3-compatible 12-6-4 parameters; state tuned chelator values are absent; treat Al/Fe distances as model-dependent geometry, not affinity evidence. Evidence: metal_parameter_values.csv; Figure TC4.	Corrected with caution
R2-Minor 1	Christopher Stuetzle spelling	Record the requested spelling correction in the response matrix only; manuscript author line is intentionally unchanged in this package. Evidence: reviewer_response_matrix.csv.	Deferred: manuscript outside scope
R2-Minor 2	Arg100 not labeled	Identify chain-A Arg100 explicitly and show when it becomes Lys100 in the Hans pocket seed. Evidence: Figure TC3; finalist_lineage.csv.	Corrected
R2-Minor 3	References 30-36 irrelevant	Record as a manuscript bibliography action; no bibliography is modified in this technical-only package. Evidence: reviewer_response_matrix.csv.	Deferred: manuscript outside scope
R2-S4 Y-distance	Y distances converge near 2.19 Å	Report all nine replica values. Similarity is due to the common Y(III) model and near-identical local coordination; displayed equality was rounding, not duplicated data. Evidence: Figure TC4; openmm_replicates.csv.	Corrected
R2-Data	Deposit raw ORCA .in/.out	Repository audit finds no ORCA/CASSCF/CASPT2 inputs or outputs. Package lists the required filenames/metadata and marks Figs. 6-7 non-regenerable. Evidence: quantum_evidence_gap_register.csv; SOURCE_DATA_GAPS.md.	Blocked: author files required

Machine-readable version: tables/reviewer_response_matrix.csv.

3. Corrected figures

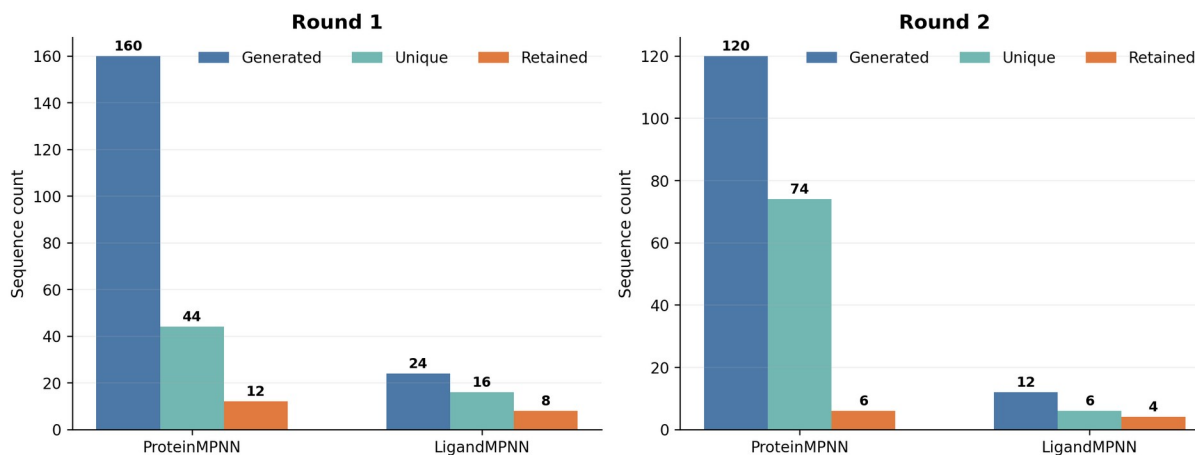
Corrected design-mask accounting



Source: results/tables/design_mask_candidates.csv. 70 is a membership sum; 66 is the unique residue inventory.

Figure TC1. Corrected design-mask accounting. Role categories overlap: 70 memberships correspond to 66 unique residue positions, with four one-count overlaps.

Corrected sequence-design funnel

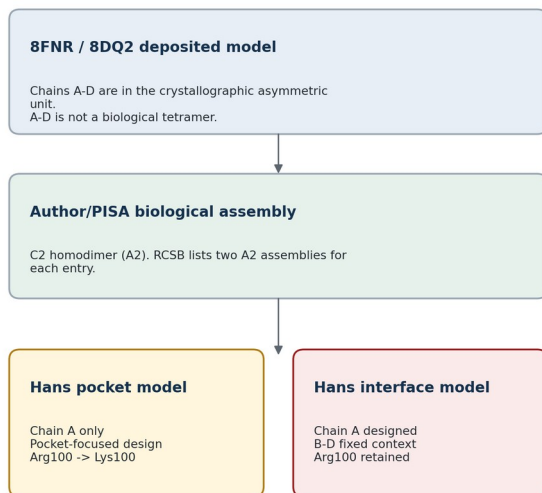


Positions, raw generations, unique sequences, and retained candidates are reported as separate units.

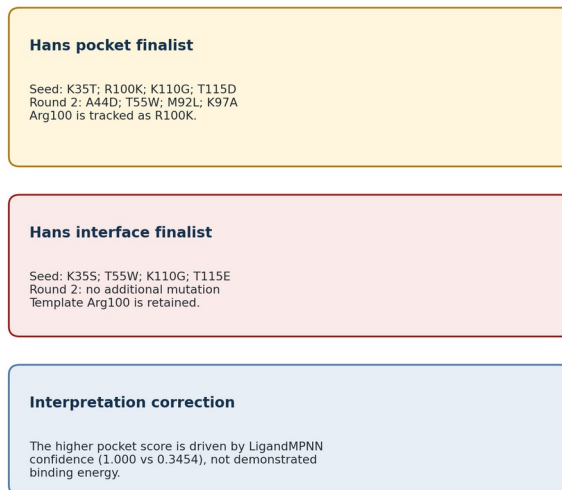
Figure TC2. Corrected design funnel. Raw generations, unique sequences, shortlisted sequences, scored candidates, and final OpenMM finalists are reported separately.

Corrected Hans model scope and Arg100 labeling

A. Structure and assembly scope



B. Unambiguous chain-A mutation lineage



This technical schematic clarifies model scope and lineage; it is not a substitute for a validated 3D contact-distance analysis.

Figure TC3. Corrected Hans model scope and Arg100 labeling. The diagram distinguishes the crystallographic A-D context from A2 biological assemblies and explicitly tracks chain-A Arg100/Lys100.

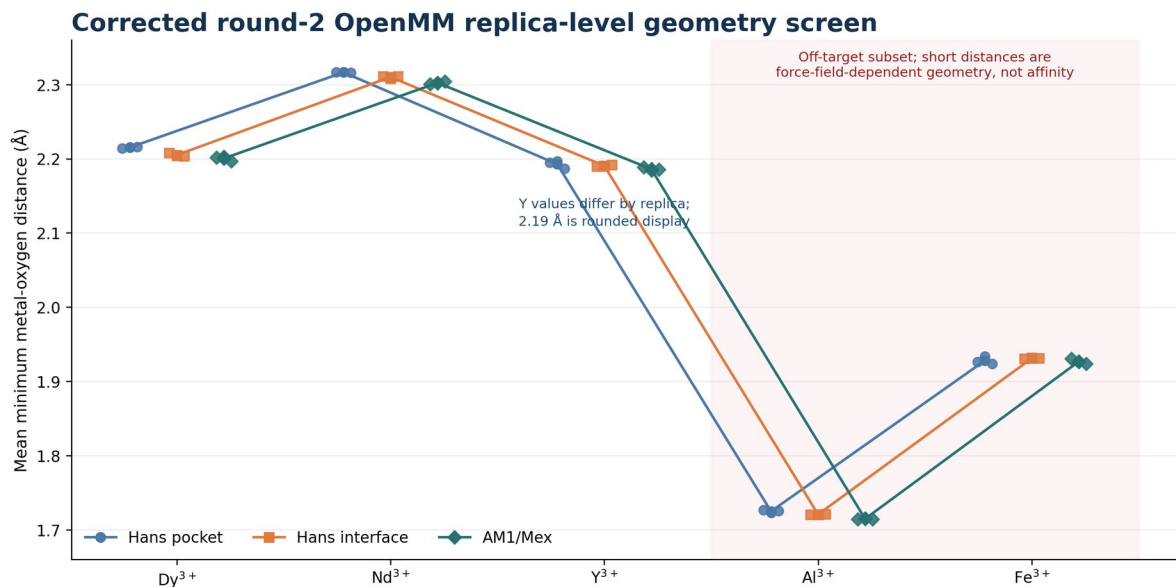
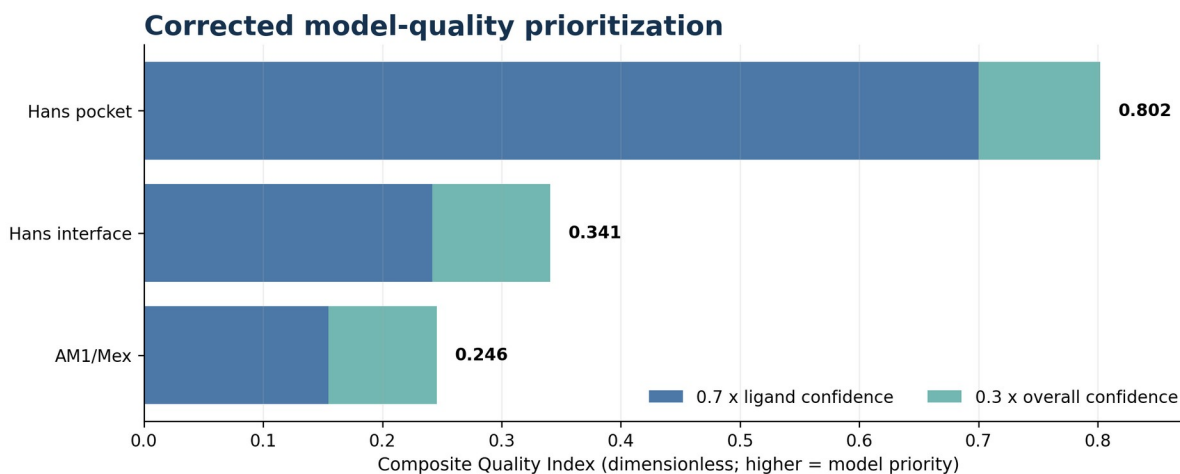


Figure TC4. Corrected round-2 OpenMM geometry screen. Every point is a replica mean. The Al/Fe values are force-field-dependent geometry results and are not interpreted as affinity or selectivity.



No RT factor and no -ln transform are used. This score is not Delta G, binding affinity, or selectivity.

Figure TC5. Corrected model-quality prioritization. The Composite Quality Index is dimensionless and must not be labeled as Delta G, free energy, or thermodynamic affinity.

4. Corrected tables and exact technical settings

4.1 Design-mask reconciliation

Role	Memberships
First-shell	26
Second-sphere	16
Interface	10
Protected	18
Unique inventory	66
Membership sum	70
Overlap excess	4

Source: design_mask_candidates.csv; overlap detail is in mask_overlaps.csv.

4.2 MPNN and Rosetta specification

Component	Exact setting	Interpretation limit
ProteinMPNN round 1	20 sequences/target at temperatures 0.10 and 0.15; seed 37; batch size 1	Generative sequence sampling; not deterministic in the mathematical sense despite fixed seed.
ProteinMPNN round 2	20 sequences/target at temperatures 0.10 and 0.15; seed 37; batch size 1	Same fixed-seed sampling settings.
LigandMPNN rounds 1/2	Temperature 0.10; seed 37; 2 designs per candidate; one batch; side-chain and ligand context enabled	Model confidence is not a probability of thermodynamic binding.
Rosetta round 2	score_jd2; nstruct=1; seed 37; fa_elec weight=0; waters retained; auto metal setup; rank within topology	Score-only triage. No relaxation, no structural-realism acceptance cutoff, and no cross-topology score comparison.

Sources: recorded run_command.txt files; rosetta_round2_candidate_ranking.csv; repository scoring code.

4.3 OpenMM protocol and metric

Component	Exact setting	Interpretation limit
OpenMM round 2	298 K; 1 atm; 2 fs; friction 1 ps ⁻¹ ; 100 ps production; 3 replicas, seeds 101-103; 500-step reporting	Short reduced rescreen; not a binding free-energy calculation.
OpenMM metric	Mean over frames of the minimum distance from each retained metal atom to any oxygen; 3.2 Å inner-sphere cutoff	Occupancy=1.0 at this cutoff does not establish selectivity or experimentally persistent binding.
Force field	amber19-all.xml + amber19/opc3.xml; generic published OPC3-compatible 12-6-4 metal family	LanM-chelator-tuned numeric parameters are absent; Al/Fe short distances require sensitivity analysis.

Sources: simulation_config.yaml, system_build_log.txt, and 45 screening_log.txt files under results/round2_openmm_screening/.

4.4 Generic OPC3-compatible 12-6-4 metal parameters

Metal	Charge	Rmin/2 (Å)	epsilon (kcal/mol)	C4 (kcal/mol Å ⁴)
Dy	3	1.632	0.0962022	183
Nd	3	1.712	0.1464093	184
Y	3	1.626	0.09289608	192
Al	3	1.361	0.01031847	363
Fe	3	1.455	0.02662782	429

Source: config/metal_parameter_values.yaml. The repository does not contain chelator-tuned numeric coefficients.

4.5 OpenMM candidate means

Candidate	Dy	Nd	Y	Al	Fe
Hans pocket	2.215 ± 0.0007	2.316 ± 0.0004	2.193 ± 0.0055	1.725 ± 0.0017	1.928 ± 0.0050
Hans interface	2.205 ± 0.0023	2.310 ± 0.0017	2.190 ± 0.0013	1.721 ± 0.0005	1.931 ± 0.0006
AM1/Mex	2.200 ± 0.0032	2.302 ± 0.0021	2.186 ± 0.0022	1.715 ± 0.0006	1.927 ± 0.0034

Values are mean ± sample SD of three replica-level means in Å. Full six-decimal data are in openmm_replicates.csv and openmm_summary.csv.

The near-identical Y values are expected from the common Y(III) parameter model and similar first-shell coordination. The underlying nine replica means range from 2.183940 to 2.196893 Å; the repeated '2.19 Å' statement was only rounded display.

4.6 Dimensionless model-quality index

Rank	Candidate	Ligand confidence	Overall confidence	CQI
1	Hans pocket	1.0000	0.3399	0.801970
2	Hans interface	0.3454	0.3303	0.340870
3	AM1/Mex	0.2213	0.3023	0.245600

CQI = 0.7 x ligand confidence + 0.3 x overall confidence. It is dimensionless and has no thermodynamic interpretation.

5. Structure and model-scope correction

PDB	Deposited model	Biological assembly	Technical use
8FNR	A,B,C,D; Dy: A=4, B=3, C=3, D=4 (14 total)	A2 homodimer (C2); two author/PISA assemblies are listed	Pocket model uses chain A; interface model keeps A-D crystallographic context
8DQ2	A,B,C,D; La: 3 per chain (12 total)	A2 homodimer (C2); author/PISA; gel-filtration evidence for assembly 1	Reference only; do not call A-D the biological tetramer
8FNS	A; Nd: 4	Monomeric reference in repository workflow	AM1/Mex chain-A model
6MI5	X; Y: 3	Monomeric reference in repository workflow	AM1/Mex reference

Primary structure sources: <https://www.rcsb.org/structure/8FNR> and <https://www.rcsb.org/structure/8DQ2>; repository inventory: [results/tables/template_chain_summary.csv](https://www.rcsb.org/results/tables/template_chain_summary.csv).

The deposited metal counts are crystallographic inventories, not numbers of equivalent functional sites. Any interface conclusion should identify the specific biological assembly or explicitly state that A-D was retained as crystallographic context.

6. Quantum evidence gap and submission gate

Submission gate

Do not present Figures 6-7, natural occupations, CASPT2 state ordering, spin-orbit conclusions, or reduced-cluster sensitivity as repository-reproduced results until the corresponding raw files are added and independently regenerated.

Required artifact	Minimum content	Status	Affected output
Dy CASSCF input and full output	Coordinates, charge/multiplicity, Hamiltonian/ECP, basis sets, CAS(9,7), roots, convergence, occupations	Absent	Dy active-space claims; Figures 6-7
Nd CASSCF input and full output	Coordinates, charge/multiplicity, Hamiltonian/ECP, basis sets, CAS(3,7), roots, convergence, occupations	Absent	Nd active-space claims; Figures 6-7
Dy/Nd CASPT2 inputs and outputs	State count, imaginary and IPEA shifts, intruder-state diagnostics, energies and convergence	Absent	CASPT2 comparison
SOC calculation files	SOC method, states mixed, spin-free roots, matrices/energies, convergence	Absent	Electronic-state ordering and anisotropy claims
Reduced-cluster coordinate and definition files	Atom/residue inventory, caps, constraints, charge/multiplicity, retained waters, cluster size	Absent	Environmental-scope claims
Natural occupation tables for every root	Seven active-orbital occupations per root with orbital identities	Absent	Active-space appropriateness
Orbital image source data	Molden/gbw or equivalent files plus orbital indices and plotting settings	Absent	Figure 7

Repository tree audit: no paths containing ORCA, CASSCF, or CASPT2 and no .in/.inp/.out quantum files were present at the audited base commit.

7. Repository correction set

Path	Purpose
README.md	Replaces the complete/reproducible and Delta-G language with a technical-correction notice and accurate limitations.
docs/technical_corrections/README.md	Human-readable scope, reviewer mapping, and regeneration instructions.
docs/technical_corrections/SOURCE_DATA_GAPS.md	Explicit quantum and manuscript-only actions that remain open.
docs/technical_corrections/LanM_Technical_Corrections_Report. {docx,pdf}	Shareable Word and PDF report.
results/technical_corrections/figures/*	Five corrected technical figures.
results/technical_corrections/tables/*	Machine-readable corrected data and provenance.
scripts/build_technical_corrections.py	Deterministic rebuild of tables, figures, report, and manifest.
requirements-technical-corrections.txt	Pinned reporting dependencies.

The manuscript file and its figures are intentionally not edited by this repository correction set.

8. Package manifest

The submission bundle includes this Word report, a PDF rendering, five corrected figures, twelve corrected/source-audit CSV tables, a figure/table manifest, repository-ready documentation, a reproducible build script, pinned reporting requirements, and SHA-256 checksums. The original reviewer PDF and manuscript are not redistributed in the generated ZIP.

Recommended share note

This technical package resolves the reproducibility, terminology, sampling, model-scope, and classical-data presentation issues that can be corrected from the repository. Quantum-source-data and manuscript-only edits remain explicit author actions.